

SEMINAR SUMMARY:
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CUTTING KNOTS DOWN TO SIZE

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The goal behind the talk was to inspect tangle decomposition as a tool for understanding 3-manifold theory and $\mathbb{R}^3$ topology. We begin by considering a few definitions.

Definition 1.

- i A *knot* is an embedding of a circle into $\mathbb{R}^3$ space or an embedding into a 3-sphere.
- ii A *link* L is union of knots.
- iii A *link diagram* is a projection of L into the plane with over-crossing and under-crossings explicitly labeled.
- iv *Spacial graphs* or a *spacial representation* $\mathcal{R}(G)$, of a graph G , is the embedded image of G in $\mathbb{R}^3$ where the vertices of G are distinct points in $\mathbb{R}^3$ and the edges between them are simple Jordan curves between them in such a way that any two curves either are disjoint or they meet in a common end point.

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- i One
- ii two
- iii three